Code Review: analysis/2025/pm_functions.R

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1 Dead Code

1.1 1. `calculate_carryover()` (lines 24-77)

Multi-model carryover function supporting exponential, linear, and Weibull decay. **Never called** by any function in the file. The actual carryover is computed inline in `apply_carryover_to_component()` using the exponential formula `(1/2)^(tsd/halflife)`. The linear and Weibull models were never integrated into the DGP.

Action: Remove entirely.

1.2 2. `calculate_component_halflives()` (lines 138-145)

Returns a list with three identical values (`br_halflife`, `pb_halflife`, `tv_halflife`), all set to `base_halflife`. Only `br_halflife` is ever used (the function guards on `component_name != "br"` in `apply_carryover_to_component()`).

Action: Inline the single value. Replace all calls with `model_param$carryover_t1half` directly.

1.3 3. `calculate_bio_response_with_interaction()` (lines 95-126)

Thin wrapper that:

1. Calls `mod_gompertz()` on `trial_data$tod`
2. Sets `bio_response_test <- base_bio_response == 0`
3. Calls `apply_carryover_to_component()`
4. Returns a 3-element list

This adds indirection without value. The orig code does this inline in `buildSigma()` in 10 lines. The `raw_bio_response_means` return field is never used by any caller.

Action: Inline into `generate_data()` and `build_sigma_matrix()`.

1.4 4. `validate_correlation_structure()` (lines 778-813)

Diagnostic function. Calls `build_sigma_matrix()` and reports eigenvalues and condition number. Not called by the simulation pipeline. Was useful during development but is now redundant with `validate_parameter_grid()`.

Action: Remove. The orig's `validateParameterGrid()` (ported as Step 7c) covers this use case.

1.5 5. `create_sigma_cache_key()` (lines 815-822)

Creates a cache key from `design_name`, `params$n_participants`, `params$biomarker_correlation`, `params$carryover_t1half`. **Never called.** The actual caching in `generate_simulated_results()` uses `paste(vpg indices, sep='_')` directly.

Action: Remove entirely.

1.6 6. `validate_parameter_grid()` (lines 828-991)

170-line validation function with extensive console output, emoji, condition number statistics, and recommendations. References `param_grid$biomarker_correlation` and `param_grid$n_participants` – column names from the old 2025 parameter format that no longer exists after the alignment. Would crash if called.

Action: Remove. The orig's `validateParameterGrid()` and the ported `generate_simulated_results()` handle pre-flight validation.

1.7 7. `report_parameter_validation()` (lines 997-1044)

Companion reporting function for `validate_parameter_grid()`. Same issue – references old column names. Never called by the pipeline.

Action: Remove.

1.8 8. Deprecated docstring block (lines 147-158)

Roxygen comment block for a function that was already removed. Describes the deprecated `calculate_carryover_adjusted_correlations()` which no longer exists.

Action: Remove.

2 Duplication

2.1 9. `generate_data()` and `build_sigma_matrix()` share ~100 lines

Both functions independently compute: labels, standard_deviations, means (including the Gompertz + carryover loop), and the correlation matrix. The logic is copy-pasted. In orig, `buildSigma()` is a standalone function and `generateData()` calls it (or uses a cached version).

Action: Make `generate_data()` always call `build_sigma_matrix()` when no cached sigma is provided, eliminating the duplicated sigma construction path (lines 433-582 of generate_data). This matches the orig architecture where `generateData()` calls `buildSigma()`.

2.2 10. `factor_types` and `factor_abbreviations` are identical

After the alignment, `factor_types = c("tv", "pb", "br")` and `factor_abbreviations = c("tv", "pb", "br")` are the same vector. Both are passed around as separate arguments and iterated in parallel. The `current_factor` / `factor_abbrev` distinction no longer exists.

Action: Collapse to a single `factors <- c("tv", "pb", "br")` throughout.

3 Simplifications

3.1 11. `apply_carryover_to_component()` over-engineered

Takes `component_halflives` (a list) and `component_name` (a string), then immediately returns unchanged if `component_name != "br"`. This function is called three times in a loop – twice it returns immediately. The function could simply accept the halflife value directly and only be called for BR.

Action: Call only for BR component. Pass `model_param$carryover_t1half` directly instead of the halflives list. Remove the `component_name` guard.

3.2 12. `prepare_trial_data()` fallback logic

Lines 238-246 handle a `week` column fallback that is never exercised. All trial designs in this codebase use `t_wk`. The `on_drug` column it adds duplicates information already available from `tod > 0`.

Action: Simplify to: add `t_wk_cumulative <- cumsum(t_wk)` and `on_drug <- (tod > 0)`. Remove the `week` fallback.

3.3 13. `build_correlation_matrix()` signature has 11 parameters

Several are vestigial: `bio_response_test`, `bio_response_means`, `means` are only used for the BM-BR correlation block, and even there `bio_response_test` is only checked when `trial_data` is NULL (which never happens in practice since `trial_data` is always passed). The `factor_types` parameter is now identical to `factor_abbreviations`.

Action: Simplify signature. Remove `bio_response_test`, `bio_response_means`, `means`, `factor_types`. Use `trial_data` (always present) for the on-drug check.

3.4 14. `process_participant_data()` pre-allocates then overwrites

Lines 1054-1058 create a zero-filled tibble and bind it, then lines 1060-1079 overwrite every column. The pre-allocation has no benefit since `mutate()` creates new columns regardless.

Action: Remove the pre-allocation block. Let the `mutate()` calls create the columns directly.

3.5 15. PD tracking in `generate_data()` is single-call accounting

`sigma_count` and `non_positive_definite_count` are initialized to 0, incremented at most once (the function builds one sigma), and then attached as attributes. The `non_positive_definite_rate` is always 0 or 1. This tracking made sense in a loop but is trivial for a single-sigma function.

Action: Replace with a single boolean attribute `was_pd_corrected`.

4 Estimated Impact

Category	Items	Lines removed	Lines simplified
Dead code	7	~350	–
Duplication	2	~100	~50
Simplifications	5	~30	~60
Total	**14**	**~480**	**~110**

Current file: 1650 lines. After cleanup: ~1060 lines.